

Assignment Three

Data Warehouse Modeling

Karen Lau (220509105)

EECS 3421: Introduction to Database Systems

Dr. Mona Nasery

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TASK 1: Literature Review

A. Source Portfolio and Credibility Assessment

BrokerLink Communications. “How to File a Car Insurance Claim in Canada.”

BrokerLink, 7 June 2025,

www.brokerlink.ca/blog/what-do-i-do-after-a-loss-a-complete-guide-to-filing-an-insurance-claim.

- **Resource Type:** Insurer; informative

- **Credibility**

This source is reliable and credible because the article was written from the perspective of the insurance company– directly from an individual in the industry. Although there is potential for bias, as the company may want to attract clients, the blog’s content is purely informational, focusing on the insurance claim process (no evident motivated or incentive-driven messaging).

Financial Consumer Agency of Canada. “Car Insurance.” *Government of Canada*, 16 Oct. 2025,

www.canada.ca/en/financial-consumer-agency/services/insurance/car.html.

- **Resource Type:** Regulator (Government); Informative

- **Credibility**

This source is credible, reliable and unbiased because it is a government agency website that provides factual information about the logistics of car insurance. It explains key topics such as what car insurance is, who is covered, the types of coverage available, and more.

McGillivray, Marisa. “Impacts of Rising Costs and Claims on Personal Automobile Insurance Profitability and Consumers in Canada.” Statistics Canada, Government of Canada, Statistics Canada, 2 Apr. 2025, www150.statcan.gc.ca/n1/pub/11-621-m/11-621-m2025003-eng.htm.

- **Resource Type:** Industry Analysis (Statistics Canada/Government)

- **Credibility**

This source is credible and reliable because it is written and published by Statistics Canada, a government agency. Its government origin also supports its objectivity; although it provides industry analysis, the article focuses strictly on numerical trends in the data, such as claim costs, claim ratios, and insurance rates by province.

TD Insurance. “How Car Insurance Premiums Are Calculated.” TD Insurance, www.tdinsurance.com/products-services/auto-car-insurance/tips-advice/how-car-insurance-premiums-are-calculated. Accessed 23 Nov. 2025.

- **Resource Type:** Insurer; Informative

- **Credibility**

This source is credible and reliable because it is sourced from one of Canada's largest multinational financial institutions and provides information on car insurance, specifically premiums. While there is potential for bias since the institution offers insurance and may seek to attract customers, the article appears to be strictly informative.

B. Conflicting Viewpoints Analysis

Below, I outline two controversial topics in the automobile insurance industry. I provide a brief summary of the opposing viewpoints for each issue and also offer my own perspective on the issue.

1. Use of Non-Driving Factors to Determine Insurance Pricing

- Many individuals argue that it is unfair for car insurance premiums to incorporate non-driving factors such as credit score, zip code, or education level. These factors are unrelated to how an individual operates their vehicle and thus, individuals believe such variables should not influence the cost of insuring a vehicle. For instance, individuals that have a lower credit score may face increased premiums. In contrast, the opposing view maintains the belief that using non-driving factors such as credit scores, is an objective way to assess applicants. They argue that these factors offer insight into a person's overall reliability and financial responsibility, which can correlate with risk.
- As a consumer, although I dislike paying higher premiums for services like car insurance, I can understand why auto insurance companies use these non-driving factors to better predict risk and price their policies accordingly. However, I also recognize that many of these non-driving factors are not easily changeable— at least in the short term. For instance, an individual may have a low credit score simply because they lack credit history or are temporarily unable to pay back a loan due to unforeseen circumstances. Similarly, individuals may live in “higher-risk” neighbourhoods because of financial limitations; this is not something they can immediately change and charging them more for insurance only worsens their situation. This creates a system that feels unfair. Perhaps, to balance both perspectives, auto insurance companies could consider these non-driving factors only during the initial months of a customer's policy. If the individual demonstrates dependable, low-risk behaviour over time, insurers should remove these factors from premium calculations. Consumers should not be penalized long-term for circumstances beyond their control.

2. Telematic Data and the Introduction of Discrimination

- As technology continues to advance, so do our automobiles. Today, most vehicles are equipped with large digital displays, built in cameras, and even self-driving capabilities. With these innovations, data has become more accessible— telematic devices can record detailed data such as acceleration, braking patterns, and other driving behaviours. The industry has begun using this data to determine car insurance premiums.

- On one hand, many individuals view this as a positive and necessary step. By basing premiums on actual driving behaviour rather than demographic factors like age or gender, insurance policies may become less prone to discrimination and bias. However, others argue that telematics can introduce new forms of unfairness. Data can be recorded inaccurately, or certain vehicles may have more additional sensors than others. In such cases, insurers may interpret the absence of certain data as the driver withholding information, when in reality the individual simply cannot afford the upgraded technology. Additionally, telematic data may misinterpret trends unrelated to driving skills. For instance, long commutes or frequent night driving is typical for rideshare drivers or night-shift workers, but may be considered risky for 'normal' individuals. If insurers rely solely on telematics without considering contextual factors, the system could introduce discrimination against these individuals.
- I also agree that using telemetric data is a step in the right direction. It reduces bias and discrimination in the pricing of insurance plans by ensuring individuals are not charged more for factors beyond their control. Instead, premiums are based solely on their driving behaviours. However, the system could be improved by integrating telematics with a limited set of relevant external factors and continuing to refine the analytical algorithms used to interpret the data. This approach may offer the best and most equitable way to price insurance plans.

C. Necessity of a Data Warehouse for Risk Analysis

A data warehouse is essential for risk analysis in auto-insurance operations because these companies rely on large volumes of data to accurately assess and identify risks among their clients. A data warehouse is perfect for this purpose for the several reasons:

- A data warehouse provides a centralized view of all data. From a database perspective, insurers do not need to join multiple tables to retrieve the data they need. Due to the structure of data warehouses, related information is stored in shared dimensions that are all connected to a central fact table. This design makes it much more convenient and efficient for insurers to access the data they need.
- Data in a data warehouse cannot be updated and is read-only, this ensures that all information is accurate and consistent. This is maintained through a meticulous extraction and transformation process that data is subjected to before being loaded into the warehouse (ETL). Additionally, since data cannot be updated or deleted, insurers have access to comprehensive historical data, allowing them to validate trends and information.
- A key aspect of risk analysis is the ability to accurately analyze potential high-risk clients or drivers based on factors such as their personal driving record, the vehicle they own, the area they live in, and broad population trends. A data warehouse can store information from both internal company records and external sources, enabling comprehensive analysis. By leveraging all of this data, insurers can make accurate predictions about new incoming clients and provide appropriate car insurance quotes.

TASK 2: Designing a Data Warehouse

A. Identification of DWH Components

I have designed a constellation schema for the data warehouse schema, which is presented as figure 1 in the [appendix](#). The schema includes two fact tables— Policy Premium and Claim Line fact tables— and 7 dimension tables— Driver, Location, Damages, Accident, Time, Car, and Coverage. Each of these tables play a role in the schema which I will discuss in more detail below:

Policy Premium Fact Table

This is one of the central tables in the constellation schema. This table records the payments that individuals make (typically monthly) to maintain their insurance coverage. These premiums are generally calculated based on a variety of factors related to risk assessment.

Grain:

- Aggregated fact table
- Coarse granularity level: Every row represents a policy per month

Numerical Measures:

- *Total Premium Amount*: Premium amount customer owes for the policy term (year)
- *Earned Premium*: Portion of the total premium that the insurer has earned so far
- *Unearned Premium*: Portion of the premium that has not been earned

This fact table is connected to 5 of the 7 dimension tables in the schema: Driver, Coverage, Car, Time, and Location.

Claim Line Fact Table

This is the other central table in the constellation schema. This table records individual components of a specific claim, where each line represents a different type of damage and its associated cost.

Grain:

- Detailed fact table
- Fine level granularity: Every row represents a *claim line*

Numerical Measures:

- *Claim Line Amount*: Records the price of the claim line
- *Paid Line Amount*: Indicates how much the insurer has paid to the claimant for the specific claim line

This table also has the primary key from the operational relational schema, claim number. This allows aggregations over claims.

This table is connected to all 7 dimension tables in the schema.

Driver Dimension Table

The driver dimension table contains information about the driver (policy holder).

Attributes:

- Driver Key (Surrogate key)
- Driver ID (PK from relational schema)

- Driver Name
- Driving Years: Years of driving experience
- Birth year: Age
- Driving Record
- Gender
- License Class
- Age Group
- Risk Category

Location Dimension Table

The location dimension table contains geographical information. When linked to the policy premium fact table, this could indicate the policy holder's home address. In contrast, in the claim line fact table, this could reference the accident location (if applicable).

Attributes:

- Location Key (Surrogate key)
- City
- Province
- Postal Code

Hierarchy:

- Postal Code → City → Province

Damages Dimension Table

The damages dimension table includes information on damages that occur during claims.

Attributes:

- Damage Key (Surrogate key)
- Damage Code (PK from relational schema)
- Damage Description
- Damage Category

Accident Dimension Table

Contains information about the accident.

Attributes:

- Accident key (Surrogate key)
- Accident type
- Accident severity

Hierarchy:

- Accident Type → Accident severity

Time Dimension Table

The time dimension provides a temporal context for the fact tables. A time entity was not provided in the original operational relational schema; however, date information can be extracted from the accident table.

Attributes:

- Time Key (Surrogate Key)
- Date

- Year
- Month
- Day of Week (e.x. Monday, Tuesday, etc.)
- Quarter
- Day
- Week of Year

Hierarchy:

- Day → Month → Quarter → Year

Car Dimension Table

The car dimension contains descriptive information about each vehicle. When linked to the policy premium fact table, the car key identifies the vehicle covered under the insurance policy. In contrast, when linked to the claims line fact table, this indicates the vehicle involved in the accident which led to the claim.

Attributes:

- Car Key (Surrogate Key)
- Car ID (PK from relational schema)
- Car Model
- Year Made
- Vehicle Type
- Value category

Hierarchy:

- Car Model → Vehicle Type → Value category

Coverage Dimension Table

The coverage dimension contains information about each of the car insurance coverage policies.

Attributes:

- Coverage Key (Surrogate Key)
- Coverage Code (PK from relational schema)
- Max Coverage
- Coverage Category

B. Design Choices

In the previous section, I provided a detailed breakdown of the fact tables, dimensions and attributes included in the constellation schema. Below I describe the rationale behind these design choices:

Fact Tables:

Policy Premium Fact Table:

This fact table is designed to support analysis of policy pricing, revenue and premium-related performance over time. The measures that are included:

- *Total premium amount*: Represents the full premium charged for the policy term (typically one year)

- Allows the business to evaluate pricing strategies and compare premium amounts between customer segments, vehicle types or geographic regions.
- *Earned Premium*: Reflects the portion of the total premium that has been earned based on how much time has passed since the policy began.
 - For instance, a policy may be worth \$1200 for a whole year; but in March (3 months passed) the earned premium would be \$300.
 - This measure is essential for monthly or period-based revenue analysis.
- *Unearned premium*: A measure that indicates the unpaid portion of the total premium.
 - Enables the company to measure and understand outstanding fees and liabilities. This also allows them to forecast future revenue.

Claim Line Fact Table:

This fact table captures detailed information for each line of a claim.

- *Claim Line amount*: Represents the amount associated with each claim line.
 - This will enable analysis of claim severity.
- *Paid line amount*: A measure that considers how much money the insurer has actually paid out.
 - Allows the business to measure realized losses, monitor their revenue and cash outflow, and assess profitability.

Dimension Tables

The dimension tables were derived from the entities in the operational relational schema. Each dimension has descriptive attributes that allow the measures in the fact table to be analyzed in a meaningful way. Entities such as driver, car, coverage, accident and damages were converted into dimensions because they contain descriptive information that is essential for trend, risk, and performance analysis. Not all attributes from these entities were kept to avoid redundancy and improve analytical clarity. In particular, certain attributes were extracted and placed into dedicated dimensions such as the location and time dimensions. These are standard dimensions in most analytical systems and enable the company to analyze temporal trends over periods of time and geographical patterns.

C. Changes

A slow changing dimension refers to a dimension in the data warehouse that changes infrequently. This is common in dimensions such as Location, where once an area is added, details such as the postal code or region are unlikely to change. In such cases, each change or new variation of a record is treated as a new entry and assigned a new surrogate key, even if it differs from an existing record by only one attribute. This approach ensures that analysis can be done on current data, while preserving historical data for reference or historical data analysis.

When late-arriving data affects dimensions, it can be handled by simply adding a new record to the corresponding dimension tables. However, late arriving data in fact tables can create issues with referential integrity, since the fact may reference a dimension that does not exist yet. To address this, a staging area is often used in the ETL process. The staging area acts as a temporary storage to hold incomplete fact records until all

necessary dimension data is available. This ensures that only complete and consistent data is loaded into the fact table.

TASK 3: Granularity, Aggregated Facts & Attribute Hierarchies

A. Granularity

The primary fact table in the constellation schema is the claim line fact table. This table has a high level of granularity (fine), as each record represents a single line on a claim, detailing an individual damage and its associated cost.

B. Aggregated Fact Table

One aggregated fact table that could be included in the schema is an aggregated claims fact table, where each record represents a single claim. This contrasts with the current claim lines fact table, in which multiple records can belong to the same claim. This aggregated table allows the insurance company to analyze how many claims occur during a given time period and how costly they are. Two key measures that could be included in this table are:

- Total claim cost: sum of total damages for the claim
- Number of damages: count of claim lines for the claim

C. Hierarchy

Two dimensions included in my constellation schema are: Time and Car. The associated 3-level attribute hierarchies for each of the dimensions are respectively:

- Month → Quarter → Year
- Car Model → Vehicle Type → Value category

D. Advantage Aggregated

An advantage of aggregated fact tables for analytical queries is improved queries performance. Aggregated fact tables contain pre-summarized information, so queries will run much faster than on high-granularity fact tables or standard operational schema tables. The necessary calculations are already stored in the aggregated fact table, whereas for the other versions, tables may need to be joined and calculations will need to be made, which can significantly increase processing time.

E. SQL Queries

Business Question: What is the total claim cost for each car model and driver age group?

OLTP Query

```
SELECT
  C.Car_Model,
  CASE
    WHEN YEAR(CURDATE()) - D.Birth_Year < 25 THEN 'Under 25'
    WHEN YEAR(CURDATE()) - D.Birth_Year BETWEEN 25 AND 40 THEN '25-40'
    WHEN YEAR(CURDATE()) - D.Birth_Year BETWEEN 41 AND 60 THEN '41-60'
    ELSE '60+'
  
```

```

END AS Age_Group,

SUM(CT.Total_Claim_Cost) AS Total_Claim_Cost
FROM
(
    SELECT
        Claim_No,
        SUM(Claim_Line_Cost) AS Total_Claim_Cost
    FROM Claim_Line
    GROUP BY Claim_No
) CT
JOIN Claim CM
    ON CT.Claim_No = CM.Claim_No
JOIN Accident A
    ON CM.Accident_ID = A.Accident_ID
JOIN Car C
    ON A.Car_ID = C.Car_ID
JOIN Insurance I
    ON C.Insurance_No = I.Insurance_No
JOIN Driver D
    ON I.Driver_ID = D.Driver_ID
GROUP BY
    C.Car_Model,
    Age_Group
ORDER BY
    C.Car_Model,
    Age_Group;

```

OLAP Query

```

SELECT
    C.Car_Model,
    D.Age_Group,
    SUM(CT.Total_Claim_Cost) AS Total_Claim_Cost
FROM
(
    SELECT
        Claim_Number,
        SUM(Claim_Line_Amount) AS Total_Claim_Cost
    FROM Claim_Line_Fact_Table
    GROUP BY Claim_Number
) CT
JOIN Claim_Line_Fact_Table F
    ON CT.Claim_number = F.Claim_number
JOIN Car C
    ON F.Car_Key = C.Car_Key
JOIN Driver D
    ON F.Driver_Key = D.Driver_Key
GROUP BY
    C.Car_Model,
    D.Age_Group
ORDER BY

```

C.Car_Model,
D.Age_Group;

The highlighted segments are nested queries which obtains the sum for each claim, since within the claim line fact table, several records can belong to the same claim.

TASK 4: Data Sources and Challenges in Building the Data Warehouse

A. Data Sources

Internal: Insurance company's internal operational system

- The primary source of data for the warehouse will come from the insurance company's operational systems. Since the data warehouse is built on the operational database relational schema, plenty of the information used to populate the warehouse— such as policies, premiums, claims, driver, and vehicle details —will be extracted directly from the internal systems.

External: Government driving records and crime reports

- Government records can be used to populate and validate attributes in the Driver Dimension, specifically the driver records attribute. Information may be gathered about historical accidents data, driving violations, or participation in driving school. We can also confirm the drivers' official license class.
- Additionally, we may store information about auto crime reports in the data warehouse to assess location-based risk through vehicle risk, vandalism, or other auto-related crimes. This will allow us to enhance the location dimension as it will help insurers understand location risk beyond the company's customer base. For instance, customers in our database may indicate that a location is low-risk because they never get targeted, but the external records may reveal that the area is a high risk neighbourhood.

B. Challenges

Creating a data warehouse is a complex and challenging task, especially because once data is loaded into the warehouse, it is immutable — it cannot be deleted or updated and exists permanently. This permanence introduces several challenges during the data loading and maintenance process. Firstly, we desire for our data warehouse to be complete and consistent. Since our data will be extracted from several data sources, it is essential to ensure that all data is formatted consistently and defined uniformly across all sources. Next, we must also find a way to deal with incomplete, conflicting, and duplicate data. Addressing these issues will require a meticulous ETL process. Moreover, finding the necessary data from external data sources may pose a great challenge in itself. These data sources are often difficult to access and rarely available for free. The company may need to pay a large sum of money over time to continue accessing new and accurate information. Finally, insurance data is highly sensitive as it contains information about an individual's demographics, address, vehicle details and financial records. It is fundamental to ensure that this information remains private through proper access controls, encryption and strict data governance policies.

TASK 5: Risk Analysis

With the data warehouse I have created, an auto insurance company will have a centralized and high-quality view of its data of policies, drivers, vehicles, coverages, claims and location. Once the company has populated the warehouse, they will be able to analyze trends across driver demographics, vehicle categories, coverage types, time periods and more. Below are several key questions and analyses that an auto insurance company can perform to better understand risk, improve pricing accuracy, and increase profitability.

1. How does location influence claim costs?

Using the Claim Line Fact Table and the connected Location Dimension, the insurer can identify which areas exhibit higher risk for accidents, vehicle thefts, vandalism, weather-related events or fraudulent claims. This will allow the company to adjust pricing by regions or postal codes, and identify high-loss regions, and potentially offer discounts to safer, low-risk areas. This will ensure that policy premiums better reflect the risk of location the driver lives in.

2. How do driver demographics influence claim costs?

By analyzing the Claim Line Fact Table and the Driver Dimension, the insurance company can evaluate the impact of different driver demographics such as age groups, gender, driving experience, driving records, license class, and risk categories. For instance the company may want to look into:

- If younger drivers submit more claims than older drivers?
- Do experienced drivers have lower severity of claim?
- Do certain license classes have higher claims?

This will allow for more accurate risk-based pricing for different demographic groups.

3. Which factors are common in severe and costly claims? What factors are common for safe drivers that rarely produce claims?

Using the Claim Line Fact Table and all of the dimension tables, the insurer can uncover trends in patterns. For instance, they may look into the following combinations:

- Driver demographics and vehicle details that produce frequent claims.
- Coverage types that are associated with high-cost claims.
- Demographics + vehicle traits that safe with low claim frequency possess.

identify the key traits/combination of factors that lead to more dangerous drivers and can price their policy accordingly.

This will allow the company to identify combinations of traits that lead to more dangerous or safer drivers and are able to adjust premiums appropriately for these customers. This will promote profitability and customer loyalty.

4. Which coverage types/plans are purchased the most frequently/bring in the most revenue?

Using the Policy Premium Fact Table, and Coverage Dimension, the insurance company can evaluate:

- Which coverage plans are purchased most frequently (which plan they are providing the most).
- Which plan produces the highest premium revenue.

- Which coverage plans may be underpriced or overpriced.
 - This analysis can only be done if the company compares the premium revenue with the claims paid (in the claim line facts table) to be able to measure the profits from each coverage type.

From these analyses, they will be able to optimize its pricing strategy and, if necessary, adjust coverage premiums for plans that are underperforming.

5. What demographics are most prevalent in contributing to premium revenue?

Using the Policy Premium Fact Table and the Driver Dimension, the insurer can determine which customer segments generate the majority of premium income. For instance, they will be able to compare the total premiums paid for:

- Young vs. Older drivers
- High-risk vs low-risk categories
- Different driving classes
- Drivers with different levels of driving experience.

With this information, the company will better understand the composition of its customer base. They will be able to detect segments to target for growth and focus on underrepresented groups to balance their risk portfolio and exposure.

Appendix

Figure 1: Data Warehouse Schema

